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A categorical approach to the semantics of argumentation

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Argumentation is a proof theoretic paradigm for reasoning under uncertainty. Whereas a 'proof' establishes its conclusion outright, an 'argument' can only lend a measure of support. Thus, the process of argumentation consists of identifying all the arguments for a particular hypothesis ϕ, and then calculating the support for ϕ from the weight attached to these individual arguments. Argumentation has been incorporated as the inference mechanism of a large scale medical expert system, the 'Oxford System of Medicine' (OSM), and it is therefore important to demonstrate that the approach is theoretically justified. This paper provides a formal semantics for the notion of argument embodied in the OSM. We present a categorical account in which arguments are the arrows of a semilattice enriched category. The axioms of a cartesian closed category are modified to give the notion of an 'evidential closed category', and we show that this provides the correct enriched setting in which to model the connectives of conjunction (&) and implication ($\Rightarrow$).
Finally, we develop a theory of 'confidence measures' over such categories, and relate this to the Dempster–Shafer theory of evidence.

1. Introduction

In almost any practical application, the design of a decision support system (DSS) involves some component of reasoning under uncertainty. This is as true in medicine as it is in many other areas where such systems are used: in business and finance, engineering and so forth. Medical decisions are rarely clear cut and usually involve some weighing up of evidence (in the case of a diagnostic aid) or costing of benefit (in a treatment planner).

There is a strong tradition of Bayesian statistics in medicine, and many of the diagnosis systems currently available are based on corresponding mechanisms of probabilistic inference. These systems have a secure theoretical foundation, but suffer serious practical problems as the intended range of the system increases. In particular, it becomes increasingly difficult to elicit reliable values for the requisite prior and conditional probabilities. There is often insufficient clinical data to calculate these directly and many of the larger systems resort to the use of 'subjective probabilities' estimated by a medical expert. A more important criticism from the point of view of practical decision making is that probabilistic systems are problem specific. They require that all the factors that may be

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relevant to a decision are identified in advance at the design stage, and hence fail to provide a flexible approach to decision support.

The 'Oxford System of Medicine' (OSM) is the prototype for a general purpose decision support tool aimed at the General Practitioner (Fox *et al.* 1990). It is intended to provide support for a variety of decision making tasks (treatment and referral as well as diagnosis) across the full breadth of medicine. Consequently, probabilistic methods have been rejected as inappropriate. Instead, the OSM employs a simple yet effective mechanism of 'reasoning by argument', which can be explained as follows. Suppose that we have a number of competing hypotheses, possibly just ϕ and $\neg\phi$, and that we want to decide between them. We might propose a number of arguments in favour of each and decide between them according to the relative strength of these arguments. In the case of medical diagnosis, we propose arguments for each of the possible diagnoses, and present the diagnosis with the greatest support as the most likely explanation of the patient's condition.

Although simple, this process of weighted argumentation is surprisingly effective. This is perhaps because it corresponds closely to the actual inference process that a doctor might use ('weighing up the pros and cons'). It is robust and performs well in situations where information is relatively sparse. Furthermore, empirical studies have shown that its performance compares favourably with that of a probabilistic system (O'Neil and Glowinski 1991), even in the restricted domains where probabilistic systems do well. This reflects the fact that, at least in medical diagnosis, the correct identification of all the relevant factors is more important than the particular weights or conditional probabilities assigned to them.

Despite the success of the 'argumentation' paradigm in a real world application, it is not clear, a priori, that it is theoretically justified. The aim of this paper is to give a precise semantics to the notion of argument embodied in the OSM, and hence demonstrate the validity of argumentation as a method of reasoning under uncertainty.

This is done axiomatically by modifying the categorical analysis of proof in intuitionistic logic. Recall that this is based on the *Curry–Howard isomorphism* (Girard *et al.* 1989), which relates the proofs of the $(\&, \Rightarrow)$ fragment of intuitionistic or minimal logic to terms of the simply typed λ-calculus, and hence to the arrows of a free cartesian closed category (Lambek and Scott 1986). In fact, this correspondence extends further so that any cartesian closed category can be viewed as a deductive system; the valid propositions being just the objects with a global element. We would like to define the notion of a 'category of arguments' so that any category that matches this definition can be viewed as a system of uncertain inference. The success of this claim depends on the provision of a general theory of the *strength of an argument* that applies to all such categories.

Arguments have the *form* of logical proof so it is reasonable to expect terms of the simply typed λ-calculus to provide an example. The crucial difference between argumentation and proof is that an argument is not necessarily expected to establish its conclusion outright. For instance, if a typed λ-term t contains free variables, these correspond to assumptions in the current context Γ and any uncertainty associated with these assumptions is inherited by the argument t. If, on the other hand, t is a closed term, it is unaffected by such uncertainty and we may regard it as a logical proof. Thus, proofs can be identified as a special kind of argument: those that are always certain. As an argument, in general, will only lend a degree of support, we are concerned with accumulation or *aggregation* of distinct arguments for the same proposition, and hence the objects of interest are finite sets of λ-terms in a particular context Γ. We note that these form the arrows of a semilattice enriched category $\mathscr{A}_\Gamma$, and this is the primary example on which we base the general theory of *evidential closed categories* in Section 2. A second example will be given by the category **Rel** of sets and relations, which is important mainly as a source of counterexamples.

As in the case of intuitionistic or minimal logic, the connectives & and $\Rightarrow$ are modelled by categorical structure on $\mathscr{A}$. This is given as extra data rather than specified via universal properties. In particular, conjunction corresponds to a tensor product on $\mathscr{A}$ with a specified comonoid structure on each object. The structure that we will describe is similar to that of a 'cartesian bicategory' (Carboni and Walters 1987) or 'cartesian sup-category' (Pitts 1988), both of which generalise the idea of a category of relations. There is also some connection with the 'dominical categories' of DiPaola and Heller (1987), which generalise categories of partial maps.

In Section 3, we will give an abstract definition of the strength of an argument in terms of a *confidence measure* on an evidential closed category. This is motivated as follows. Imagine a dialogue in which the first person proposes arguments and the second person either accepts or rejects them according to whether or not he or she finds them convincing. The strength of argument required will vary according to the standards and beliefs of the second person. For example, a logician will only accept logical proof, whereas a more trusting individual might tolerate nonlogical steps providing that the argument is otherwise well formed. We can measure our *confidence* in a particular argument in various ways. We might, for example, look at the set of people who would accept it, or at the time taken to convince a particular person, or simply rate it on a scale of 0 to 10. These measures may look quite ad hoc, but we will see that they all share the same basic mathematical properties.

1.1. *A rational reconstruction of the OSM*

The following example of diagnosis and treatment illustrates the key features of argumentation and the way in which it is used by the OSM.

Example 1. Suppose that we were to advise a doctor on the possible treatment of a patient who is complaining of muscular aches and pains. Aspirin is an effective remedy for muscle pain, and we might cautiously propose this as an argument for prescribing aspirin. The reason for caution is that aspirin cannot be taken by anyone suffering from a gastric ulcer. If there were any evidence to suggest that the patient might have a gastric ulcer, if he mentioned stomach pain after meals for instance, this would be a strong argument against prescribing aspirin. The doctor will weigh up the arguments presented and presumably reject the hypothesis of prescribing asprin to a patient suffering from both muscle pain and stomach pain.

On the other hand, if the patient is known to be going through a period of stress or anxiety, this may account for any stomach pain. Provided that all the symptoms have been sufficiently short term, we could argue against the presence of a gastric ulcer. This, in turn, would give us a much stronger argument for the prescription of aspirin and the hypothesis might now be accepted.

The machine solution of this problem uses the following set of facts, which would be taken from some large context Γ representing the database of medical knowledge available to us.

t_1	:	$\mathtt{muscle\_pain} \Rightarrow \mathtt{aspirin}$	0.4
t_2	:	$\mathtt{muscle\_pain} \,\&\, \mathtt{not\_gastric\_ulcer} \Rightarrow \mathtt{aspirin}$	0.6
c_1	:	$\mathtt{gastric\_ulcer} \Rightarrow \mathtt{not\_aspirin}$	1.0
i_1	:	$\mathtt{stomach\_pain} \Rightarrow \mathtt{gastric\_ulcer}$	0.5
i_2	:	$\mathtt{short\_term} \,\&\, \mathtt{anxiety} \Rightarrow \mathtt{not\_gastric\_ulcer}$	0.6

The arguments to support a particular proposition can be produced automatically by a suitable 'argumentation theorem prover' (ATP). The particular one that has been developed as part of the Praxis project is a Prolog implementation of a simple backward chaining algorithm. The arguments are returned as λ-terms using the de Bruijn notation for bound variables. The degree of support that these lend to the proposition depends on the atomic steps that they use. Each of the facts above has been assigned a strength, or confidence, in the interval $[0, 1]$. We will give several ways to assess the strength of a chain of argument, but the simplest is to take the strength of its weakest link. Adopting this rather simple scheme, we obtain the following dialogue with the ATP.

```
argue(muscle_pain & stomach_pain => aspirin, Args, Conf).

Args = [\? apply(t1,fst(0))]
Conf = 0.4

argue(muscle_pain & stomach_pain => not_aspirin, Args, Conf).

Args = [\? apply(c1,apply(i1,snd(0)))]
Conf = 0.5
```

Thus, if the patient complains of muscle pain and stomach pain and we have no further evidence, we would not recommend aspirin. The argument against carries greater weight.

```
argue(muscle_pain & stomach_pain &
       short_term & anxiety => aspirin, Args, Conf).

Args = [\? apply(t1,fst(0)),
        \? apply(t2,pair(fst(0),apply(i2,snd(snd(0)))))]
Conf = 0.6

argue(muscle_pain & stomach_pain &
       short_term & anxiety => not_aspirin, Args, Conf).

Args = [\? apply(c1,apply(i1,fst(snd(0))))]
Conf = 0.5
```

On the other hand, given additional evidence to show that the patient does not have a gastric ulcer, we obtain a stronger argument in favour of aspirin and the decision would be reversed.

There are two main points to note here. The first is the role of contradiction. The evidence supplied by a patient will often contain conflicting information. Nevertheless, we must continue to reason effectively in the face of an apparent contradiction. The approach taken here is to treat contradiction at the meta-level rather than within the logic. Thus, the notion of negation is not explicit in our logic. The propositions `aspirin` and `not_aspirin` are competing hypotheses but there is nothing to suggest that they are either complementary or exclusive. If there are arguments for both, we can resolve the contradiction by making a *decision* at the meta-level. However, we might just as well leave it unresolved as in the case of `gastric_ulcer` verses `not_gastric_ulcer` above. The issues of treating contradiction at the meta-level are explored more thoughly in other papers (Fox, Krause and Ambler 1992; Krause *et al.* 1995). Here we will merely accept it as a convenient simplification.

The second point is the the method of aggregation. In the scheme adopted here, if there are several arguments for the same proposition ϕ, the support for ϕ is taken to be the strength of the strongest one; the presence of additional arguments has no effect. Later on, in Section 3.1, we will introduce a probabilistic notion of strength. This has the property that independent arguments will mutually reinforce each other so that their combined strength is greater than that of either one alone. Under this latter scheme the support for `aspirin` would rise to 0.616 (the two arguments are assumed to be indendent – t_2 represents the *extra* force of argument that we can use if the patient is known not to have a gastric ulcer).

Note, however, that the aggregation of arguments is necessarily monotone. To overturn a decision one must supply a stronger argument on the other side. There is no mechanism for one argument to *defeat* or undercut another. Thus, the argumentation approach to default reasoning is to pose default arguments with sufficient caution that they might later be overruled.

1.2. *The syntactic category* $\mathscr{C}_\Gamma$

To construct a category $\mathscr{A}_\Gamma$ whose arrows correspond to finite sets of λ-terms in the context Γ, we first construct a category $\mathscr{C}_\Gamma$ whose arrows correspond to single λ-terms.

Let $\mathscr{L}$ denote the simply typed λ-calculus with pairing. We recall (Girard *et al.* 1989) that the Curry–Howard isomorphism demonstrates an equivalence between natural deduction proofs in the $(\&, \Rightarrow)$ fragment of minimal logic and terms of $\mathscr{L}$ such that proof normalisation corresponds to $\beta\eta$ reduction.

There is a well-known construction (Lambek and Scott 1986) that relates a syntactic category $C(\mathscr{T})$ to a type theory $\mathscr{T}$. Applying this to $\mathscr{L}$, we obtain something that would be a cartesian closed category but for the absence of a terminal object. Adjoining a unit type to $\mathscr{L}$ in the most obvious way is problematic because the required reduction rules fail to satisfy the Church–Rosser property. If we were to modify the theorem prover in this way, the proof terms would no longer have a unique normal form. We avoid the problem by treating the unit type as a special case.

Let $\mathscr{L}_1$ denote the following extension of $\mathscr{L}$. The types of $\mathscr{L}_1$ are those of $\mathscr{L}$ together with a unit type $\mathbb{1}$, and the terms are those of $\mathscr{L}$ together with a special variable $*$. We extend the type inference relation of $\mathscr{L}$ by the following three rules.

$$\frac{\Gamma \vdash t : B}{\Gamma \vdash_1 t : B} \qquad \frac{\Gamma \vdash t : B}{\Gamma, * : \mathbb{1} \vdash_1 t : B} \qquad \frac{}{\Gamma \vdash_1 * : \mathbb{1}} \tag{1}$$

Note that $*$ is the only term of type $\mathbb{1}$ and that it never appears as a subterm of anything but itself. There is therefore no need to introduce reduction rules for the unit type.

If Γ is a context in $\mathscr{L}$, we define a syntactic category $\mathscr{C}_\Gamma$ as follows. The objects of $\mathscr{C}_\Gamma$ are the types of $\mathscr{L}_1$, and the morphisms from A to B are given by equivalence classes of pairs $\langle x, t\rangle$, where x is a variable and t is a term such that

$$\Gamma, x : A \vdash_1 t : B \tag{2}$$

is derivable in $\mathscr{L}_1$ under the equivalence relation defined by $\beta\eta$ conversion and change of bound variables (including x). Let $(x \mapsto t)$ denote the equivalence class containing $\langle x, t\rangle$. Composition is given by substitution

$$(y \mapsto u)(x \mapsto t) = (x \mapsto u[t/y]) \tag{3}$$

and identities are maps of the form $(x \mapsto x)$. It is routine to verify that this data defines a category.

Proposition 2. $\mathscr{C}_\Gamma$ is a cartesian closed category.

Proof. The cartesian product and internal hom are given by extending the type building operations $\times$ and $\Rightarrow$ by the following definitions,

$$A \times \mathbb{1} = A = \mathbb{1} \times A \qquad (\mathbb{1} \Rightarrow A) = A \qquad (A \Rightarrow \mathbb{1}) = \mathbb{1}. \tag{4}$$

That is, the usual canonical isomorphisms are supressed. We define the various other data accordingly. The projections $\pi_{B,C} : B \times C \to B$ and $\pi'_{B,C} : B \times C \to C$ are defined by

$$\pi_{B,C} = \begin{cases} (y \mapsto *) & B = \mathbb{1} \\ 1_B & C = \mathbb{1} \\ (z \mapsto \mathit{fst}(z)) & \text{otherwise} \end{cases} \qquad \pi'_{B,C} = \begin{cases} 1_C & B = \mathbb{1} \\ (x \mapsto *) & C = \mathbb{1} \\ (z \mapsto \mathit{snd}(z)) & \text{otherwise,} \end{cases} \tag{5}$$

and the pairing of maps $(x \mapsto s) : A \to B$ and $(x \mapsto t) : A \to C$ is given by

$$\langle (x \mapsto s), (x \mapsto t)\rangle = \begin{cases} (x \mapsto \langle s, t\rangle) & B \neq \mathbb{1} \neq C \\ (x \mapsto s) & B \neq \mathbb{1} = C \\ (x \mapsto t) & B = \mathbb{1} \neq C \\ (x \mapsto *) & B = \mathbb{1} = C. \end{cases} \tag{6}$$

The evaluation maps $\varepsilon_{B,C} : (B \Rightarrow C) \times B \to C$ are defined by

$$\varepsilon_{B,C} = \begin{cases} 1_C & B = \mathbb{1} \\ (x \mapsto *) & C = \mathbb{1} \\ (y \mapsto \mathit{fst}(y)(\mathit{snd}(y))) & \text{otherwise,} \end{cases} \tag{7}$$

and the currying of a map $(x \mapsto t) : A \times B \to C$ is given by

$$\Lambda(x \mapsto t) = \begin{cases} (y \mapsto \lambda z.t[\langle y, z\rangle/x]) & A \neq \mathbf{1} \neq C, B \neq \mathbf{1} \\ (* \mapsto \lambda x.t) & A = \mathbf{1} \neq C, B \neq \mathbf{1} \\ (y \mapsto *) & A \neq \mathbf{1} = C, B \neq \mathbf{1} \\ (* \mapsto *) & A = \mathbf{1} = C, B \neq \mathbf{1} \\ (x \mapsto t) & B = \mathbf{1}. \end{cases} \tag{8}$$

We need to check that these data satisfy the equations defining a cartesian closed category: this is routine, though tedious. □

2. Evidential categories

In this section, we examine the structure required for a category $\mathscr{A}$ to be regarded as a category of propositions and arguments. The main example is a category $\mathscr{A}_\Gamma$, which has the same objects as $\mathscr{C}_\Gamma$ but whose morphisms $f \in \mathscr{A}_\Gamma(A, B)$ are finite sets of morphisms in $\mathscr{C}_\Gamma(A, B)$. Though the structure that we propose is quite abstract and derived from first principles, it nevertheless captures the important features of this example.

It is useful to represent the process of accumulating or *aggregating* arguments as an explicit operation on the hom-sets; in fact, this is central to our treatment. If f and g are arguments from A to B, their aggregate $f \vee g : A \to B$ is the argument whose content is precisely that of f and g considered separately. That is, $f \vee g$ is the argument that says:

"I have two reasons for believing that B follows from A,
and they are 'f' and 'g'."

We also introduce the idea of a vacuous argument from A to B. This is the argument 0 that has no content and that no-one should accept. In the case of $\mathscr{A}_\Gamma$, the aggregation of two finite subsets of $\mathscr{C}_\Gamma(A, B)$ is just their union and the vacuous argument from A to B is the empty set. It seems clear that, in a general category of arguments $\mathscr{A}$, aggregation should have the elementary properties of union, so that arguments from A to B form a join semilattice $\langle \mathscr{A}(A, B), \vee, 0\rangle$

Consider the following arrows in $\mathscr{A}$.

$$\begin{tikzcd} A \arrow[r, "k"] & B \arrow[r, bend left=60, "f \vee g"] \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] \arrow[r, bend right=60, "0"'] & C \arrow[r, "h"] & D \end{tikzcd}$$

We need to ascribe a meaning to the various composite arrows. For instance, the only sensible way to interpret $h(f \vee g)$ as an argument is to equate it with $hf \vee hg$. This is borne out by considering composition in $\mathscr{A}_\Gamma$, where the composite of finite sets $f \subseteq \mathscr{C}_\Gamma(A, B)$ and $g \subseteq \mathscr{C}_\Gamma(B, C)$ is the set of composites $\{\beta\alpha \mid \alpha \in f, \beta \in g\}$. We therefore require that composition distributes over the semilattice structure of the hom-sets:

$$\begin{array}{cc} h(f \vee g) = hf \vee hg & (f \vee g)k = fk \vee gk \\ h0 = 0 & 0k = 0, \end{array} \tag{9}$$

and hence that $\mathscr{A}$ is an **SLat**-enriched category. We will attempt to formulate the semantics of argumentation, as far as possible, within the theory of **SLat**-enriched categories. This provides a much needed guiding principle, which directs the rest of the theory.

2.1. *Conjunction*

We now consider the categorical structure needed on an **SLat**-category $\mathscr{A}$ to represent conjunction. In the case of ordinary categories, this would be given by a cartesian product, but in the context of **SLat**-categories, this view is too simplistic because finite products coincide with their corresponding coproducts so conjunction becomes confused with disjunction. We therefore seek to modify the standard account.

It is natural to require that the conjunction of arguments distributes over aggregation in much the same way as their composition. Thus, conjunction corresponds to an **SLat**-functor $\otimes : \mathscr{A}\hat{\otimes}\mathscr{A} \to \mathscr{A}$, where $\hat{\otimes}$ denotes the tensor product of **SLat**-categories. The associativity, commutativity, and unit laws of conjunction are expressed via natural isomorphisms satisfying the usual coherence conditions (Kelly 1982). The structure so far described is a symmetric monoidal **SLat**-category, and hence corresponds to a simple form of linear logic (Girard 1987). To capture the full strength of conjunction, we require that each object comes equipped with a specified comonoid structure that provides arguments to duplicate and eliminate hypotheses.

We will call the functor $\otimes$ the *evidential product* on $\mathscr{A}$. Intuitively, an arrow into A provides evidence for A, and hence an arrow into $A \otimes B$ is the conjunction of evidence for A and evidence for B.

Definition 3. An **SLat**-category $\mathscr{A}$ is *evidential* if there exists an **SLat**-functor

$$\otimes : \mathscr{A}\hat{\otimes}\mathscr{A} \to \mathscr{A}$$

and object I of $\mathscr{A}$, together with isomorphisms

$$\begin{aligned} a_{X,Y,Z} &: (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z) \\ c_{X,Y} &: X \otimes Y \to Y \otimes X \\ r_X &: X \otimes I \to X, \end{aligned}$$

and morphisms

$$\begin{aligned} \Delta_X &: X \to X \otimes X \\ t_X &: X \to I \end{aligned}$$

such that

1 the underlying category $\langle \mathscr{A}_o, \otimes_o, I, a, c, r \rangle$ is symmetric monoidal,
2 for each object X of $\mathscr{A}$, the structure $C(X) = \langle X, \Delta_X, t_X \rangle$ is a commutative comonoid in $\mathscr{A}$,

3 $C(I) = \langle I, r_I^{-1}, 1_I \rangle$ and $C(X \otimes Y) = \langle X \otimes Y, m(\Delta_X \otimes \Delta_Y), r_I(t_X \otimes t_Y) \rangle$ where m is the 'middle four interchange',
4 each morphism $f : X \to Y$ in $\mathscr{A}$ is a lax homomorphism of comonoids $C(X) \to C(Y)$, that is, the following inequalities hold

$$\begin{tikzcd} I \arrow[d, "1"'] & X \arrow[l, "t_X"'] \arrow[r, "\Delta_X"] \arrow[d, "f"] \arrow[l, phantom, "\geq", shift right=20] & X \otimes X \arrow[d, "f \otimes f"] \arrow[l, phantom, "\leq", shift right=20] \\ I & Y \arrow[l, "t_Y"] \arrow[r, "\Delta_Y"'] & Y \otimes Y \end{tikzcd} \tag{10}$$

This definition should be compared with that of a 'cartesian bicategory' given in Carboni and Walters (1987) and the related idea of a 'cartesian sup-category' given in Pitts (1988). The main difference being that we drop the requirement that the maps t_X and Δ_X have right adjoints. We make up for this loss with the inclusion of Condition 3, which relates the comonoid structure on $\mathscr{A}$ to the tensor product. It says that the comonoid structure on I is the trivial one and that the comonoid structure on $X \otimes Y$ is the tensor product of those on X and Y.

Given diagram (10), it is tempting to think that the morphisms t_X and Δ_X are the components of corresponding lax natural transformations. However, in neither case is the codomain an **SLat**-functor, so such a description would lie outside the framework of **SLat**-enriched category theory. It is therefore not clear whether the comonoid structure on $\mathscr{A}$ can be related to an adjunction in the manner of Carboni *et al.* (1990).

Example 4. Every **CSLat**-category that is cartesian in the sense of Pitts (1988) is also an evidential category. In particular, the category **Rel** of sets and relations is an evidential category with $X \otimes Y = \{\langle x, y \rangle \mid x \in X, y \in Y\}$ and $I = \{*\}$.

Example 5. Given any category $\mathscr{C}$, we can define an **SLat**-category $\wp_{\mathrm{fin}}(\mathscr{C})$ as follows: the objects of $\wp_{\mathrm{fin}}(\mathscr{C})$ are those of $\mathscr{C}$; morphisms in $\wp_{\mathrm{fin}}(\mathscr{C})(A, B)$ are finite sets of morphisms in $\mathscr{C}(A, B)$; identities are the singletons $\{1_A\}$; and composition is given by the multiplication of 'complexes'

$$\{\alpha_i\}_{1 \leq i \leq n} \{\beta_j\}_{1 \leq j \leq m} = \{\alpha_i \beta_j\}_{1 \leq i \leq n, 1 \leq j \leq m}. \tag{11}$$

In fact, $\wp_{\mathrm{fin}}(\mathscr{C})$ is the free **SLat**-category generated by $\mathscr{C}$. If $\mathscr{C}$ has finite products, $\wp_{\mathrm{fin}}(\mathscr{C})$ is an evidential category with $\otimes$ given by

$$\{\alpha_i\}_{1 \leq i \leq n} \otimes \{\beta_j\}_{1 \leq j \leq m} = \{\alpha_i \times \beta_j\}_{1 \leq i \leq n, 1 \leq j \leq m} \tag{12}$$

and a, c, r, Δ, t given by the evident singletons.

In particular, let $\mathscr{C}_\Gamma$ be the syntactic category generated by the context Γ. Then $\mathscr{A}_\Gamma = \wp_{\mathrm{fin}}(\mathscr{C}_\Gamma)$ is the evidential category whose objects are types and whose morphisms $A \to B$ are finite sets of equivalence classes $(x \mapsto t) \in \mathscr{C}_\Gamma$. We adopt the following notation for arrows in $\mathscr{A}_\Gamma$.

$$(x \mapsto t_1, \ldots, t_n) = \{(x \mapsto t_i) \mid 1 \leq i \leq n\}. \tag{13}$$

Example 6. Let D be a distributive lattice and let $\mathscr{D}$ denote the corresponding one object **SLat**-category with $\mathscr{D}(*,*) = D$, composition given by meet and $1_* = \top$. Then $\mathscr{D}$ is an evidential category in which all the distinguished maps are equal to $\top$ and the tensor product of maps is also given by meet.
In fact, any one object evidential category is of this form, since we can show that $t_* = \Delta_* = r_* = 1_*$, and hence by diagram (10) the tensor product is a meet. It coincides with composition, since $f \otimes g = (f \otimes 1)(1 \otimes g) = fg$.

2.2. *Selected Maps*

The notion of an evidential category should be regarded as a suitable translation of that of a 'category with finite products' from the context of ordinary categories to that of **SLat**-categories. In this section, we will see that the evidential product retains some of the properties of a cartesian product and show that we can recover a cartesian product from it by restricting to a particular class of 'selected' morphisms.

Given an evidential category $\mathscr{A}$, we can define 'projections' for the evidential product:

$$\begin{array}{rcl} p_{X,Y} & = & r_X(1_X \otimes t_Y) \\ q_{X,Y} & = & l_Y(t_X \otimes 1_Y), \end{array} \tag{14}$$

where l_Y is the isomorphism $r_Y c_{I,Y} : I \otimes Y \to Y$. We can also define the 'pairing' $\langle f,g\rangle : X \to Y \otimes Z$ of morphisms $f : X \to Y$ and $g : X \to Z$ via

$$\langle f,g\rangle = (f \otimes g)\Delta_X, \tag{15}$$

As $\otimes$ is bilinear, these cannot satisfy the usual equations defining a cartesian product. For instance, $p\langle f,0\rangle = 0$ regardless of f.

Lemma 7. The following inequalities hold in any evidential category:

$$p\langle f,g\rangle \leq f \tag{16}$$

$$q\langle f,g\rangle \leq g \tag{17}$$

$$h \leq \langle ph,qh\rangle. \tag{18}$$

Proof. We can show (16) as follows:

$$p\langle f,g\rangle = r(f \otimes tg)\Delta = fr(1 \otimes tg)\Delta \leq fr(1 \otimes t)\Delta = f, \tag{19}$$

and similarly for (17). To verify (18), we observe that

$$\begin{array}{rcl} (p \otimes q)\Delta_{X\otimes Y} & = & (r \otimes l)((1 \otimes t_Y) \otimes (t_X \otimes 1))m(\Delta_X \otimes \Delta_Y) \\ & = & (r \otimes l)m((1 \otimes t_X)\Delta_X \otimes (t_Y \otimes 1)\Delta_Y) \\ & = & (r \otimes l)m(r^{-1} \otimes l^{-1}), \end{array} \tag{20}$$

which, by coherence, is the identity on $X \otimes Y$. It follows that

$$h = (p \otimes q)\Delta h \leq (p \otimes q)(h \otimes h)\Delta = \langle ph,qh\rangle. \tag{21}$$

□

It is reasonable to ask for what class of maps do the converse of these inequalities hold.

Definition 8. A morphism $f : X \to Y$ in an evidential category $\mathscr{A}$ is said to be *simple* if the right-hand side of diagram (10) commutes exactly (that is, $\Delta_Y f = (f \otimes f)\Delta_X$), and *entire* if the left-hand side commutes exactly (that is, $t_X = t_Y f$). It is said to be *selected* if it is both simple and entire, that is, if it is a strict homomorphism of comonoids.

Example 9. If $\mathscr{C}$ is a category with finite products, then $f \in \wp_{\text{fin}}(\mathscr{C})(A, B)$ is simple if it contains at most one element and entire if it contains at least one element. Thus, the selected maps in $\wp_{\text{fin}}(\mathscr{C})$ are the singletons.

Example 10. A relation $F \subseteq X \times Y$ is simple if and only if it is single-valued (that is, xFy and xFy' implies $y = y'$) and entire if and only if it is total (that is, for all $x \in X$ there exists a y such that xFy). Thus, the selected maps in **Rel** are the functions.

It is clear from the proof of Lemma 7 that $p\langle f, g\rangle = f$ provided that g is entire, and $h = \langle ph, qh\rangle$ provided that h is simple. So, in particular, the inequalities (16) to (18) become equalities when all the maps are selected.

Lemma 11.

1 Every morphism with a right adjoint is selected; in particular, every isomorphism is selected.
2 The morphisms t_X and Δ_X are selected for all X.
3 The composition and tensor product of selected maps are selected.

Proof.

1 Let f^* be right adjoint to $f : X \to Y$. Then $t_X \leq t_X f^* f \leq t_Y f$ and $(f \otimes f)\Delta_X \leq (f \otimes f)\Delta_X f^* f \leq (ff^* \otimes ff^*)\Delta_X f \leq \Delta_X f$. It follows that f is selected, since the converse inequalities hold by definition.
2 The commutativity of the appropriate squares can be verified from the data given in Condition 3 of Definition 3. There is a sizeable diagram chase required to check that $\Delta_{X \otimes X}\Delta_X = (\Delta_X \otimes \Delta_X)\Delta_X$, but this is otherwise routine.
3 That selected maps are closed under composition is trivial, and closure under tensor product follows easily from Condition 3 of Definition 3. □

We note that in **Rel**, the selected maps are precisely the ones with right adjoints. However, this property is specific to **Rel** and does not hold, for example, in $\wp_{\text{fin}}(\mathscr{C})$.

The following result relates an evidential product to a cartesian one.

Proposition 12. Let $\langle \mathscr{A}, \otimes, I\rangle$ be an evidential category and let $\mathscr{S}$ be the subcategory of the underlying category $\mathscr{A}_0$ whose objects are those of $\mathscr{A}$ and whose morphisms are selected maps. Then I is a terminal object in $\mathscr{S}$ and the tensor product $\otimes$ restricts to a cartesian product on $\mathscr{S}$ in such a way that the coherence maps of $\otimes$ become the canonical coherences of a cartesian product.

Proof. That I is terminal in $\mathscr{S}$ follows immediately from the fact that selected maps are entire.

By Lemma 11, $\mathscr{S}$ contains the projections $p_{X,Y}$ and $q_{X,Y}$ for all X and Y, and is closed under pairing. As we have seen, the inequalities of Lemma 7 become equalities for selected maps, and hence this data defines a cartesian product on $\mathscr{S}$.

It is now easy to check that the coherences of $\otimes$ can be expressed in terms of the limit structure on $\mathscr{S}$: for example, we may calculate directly from the definitions that

$$a\langle\langle p, pq\rangle, qq\rangle = \langle p, \langle pq, qq\rangle\rangle = 1, \tag{22}$$

so $a^{-1} = \langle\langle p, pq\rangle, qq\rangle$ as one might expect. □

Unfortunately, the tensor product is not fully determined by the class of selected maps, since there is, in general, no way to extend from a cartesian product on $\mathscr{S}$ to a tensor product on the whole of $\mathscr{A}$.

Proposition 12 is closely related to a result by Fox (Fox 1976), which says that if $\langle \mathscr{C}, \otimes, I\rangle$ is a symmetric monoidal (bi)category, then $\otimes$ is a cartesian (bi)product on $\mathscr{C}$ and I is terminal in $\mathscr{C}$ if and only if every object has a unique commutative comonoid structure and every morphism is a homorphism of comonoids.

2.3. *Implication*

Having specified the categorical structure that represents conjunction, we can now relate this to implication in the usual way.

Definition 13. An evidential category $\mathscr{A}$ is said to be *closed* if for each object X the **SLat**-functor $(-) \otimes X$ has a specified right adjoint $[X, -]$.

We note from Kelly (1982, page 50) that $(-) \otimes X$ has a right adjoint in **SLat-Cat** provided that its underlying functor has an adjoint in **Cat**. The adjunction gives rise to an isomorphism of semilattices

$$\Lambda : \mathscr{A}(A \otimes X, B) \cong \mathscr{A}(A, [X, B]), \tag{23}$$

which is natural in A and B. We call the transpose $\Lambda(f)$ of a morphism f the *currying* of f, and refer to the components $\varepsilon_{X,Y} : [X, Y] \otimes X \to Y$ of the counit as *evaluation maps.* As in the case of ordinary categories, the functors $[X, -]$ may be pieced together to give a single **SLat**-functor $[-, -] : \mathscr{A}^{op}\hat{\otimes}\mathscr{A} \to \mathscr{A}$ called the *internal hom* of $\mathscr{A}$.

The laxness of diagram (10) is to some extent a prerequisite for $\mathscr{A}$ being closed. If the left-hand square were to commute exactly for all f, then, taking $f = 0 : I \to I$, we see that I must be a zero object, and hence $\mathscr{A}$ could not be closed without being degenerate via the isomorphism $\mathscr{A}(X, Y) \cong \mathscr{A}(I, [X, Y])$.

Example 14. If $\mathscr{C}$ is a cartesian closed category, $\wp_{\text{fin}}(\mathscr{C})$ is evidential closed with the internal hom given on objects by $[X, Y] = X \Rightarrow Y$, evaluation given by the singleton $\{\varepsilon\}$, and the currying of a set defined

$$\Lambda\{\alpha_i\}_{1\le i\le n} = \{\Lambda\alpha_i\}_{1\le i\le n}. \tag{24}$$

Example 15. The category **Rel** of sets and relations is evidential closed with $[X, Y]$ equal to $X \otimes Y$, evaluation given by the relation $E = \{\langle\langle\langle x, y\rangle, x\rangle, y\rangle \mid x \in X, y \in Y\}$ and the currying of a relation $F : X \otimes Y \to Z$ defined by

$$\Lambda(F) = \{\langle x, \langle y, z\rangle\rangle \mid \langle\langle x, y\rangle, z\rangle \in F\}. \tag{25}$$

Example 16. Let D be a distributive lattice and let $L = \mathbf{Idl}(D)$ be the ideals of D ordered by inclusion. We recall (Johnstone 1982) that L is a locale, and that the finite

elements of L are precisely the principal ideals of D. Let A be any subset of the ideals in D containing $\top$ and closed under $\wedge$ and $\rightarrow$. For each pair $x, y \in A$, we define $\mathscr{A}(x,y) = \{d \in D \mid d\!\downarrow\, \leq x \rightarrow y\}$. As the principal ideals are closed under finite join, we have that $\mathscr{A}(x,y)$ is a join semilattice. Furthermore, $\top \in \mathscr{A}(x,x)$ for all $x \in D$ and if $d_1 \in \mathscr{A}(x,y)$ and $d_2 \in \mathscr{A}(y,z)$, then clearly $d_1 \wedge d_2 \in \mathscr{A}(x,z)$. We can therefore define the **SLat**-category $\mathscr{A}$ whose objects are the elements of A and whose morphisms are given by the hom-objects $\mathscr{A}(x,y)$ such that composition is given by $\wedge$ and the identities are given by $\top$.
We note that the operations $\wedge$ and $\rightarrow$ on A extend to **SLat**-functors $\otimes : \mathscr{A}\hat{\otimes}\mathscr{A} \rightarrow \mathscr{A}$ and $[-,-] : \mathscr{A}^{op}\hat{\otimes}\mathscr{A} \rightarrow \mathscr{A}$ whose action on morphisms is given by $\wedge$ in both cases. It is now easy to verify that $\mathscr{A}$ forms an evidential closed category in which all of the distinguished morphisms are given by $\top$.

2.4. *Logical Maps*

A logical proof is just a particular form of argument, and we pick out the arrows of $\mathscr{A}$ that correspond to proofs via the following definition.

Definition 17. If $\mathscr{A}$ is an evidential closed category, the class of *logical maps* is the smallest class that contains the identities, the arrows l_X, r_X, Δ_X, t_X and $\varepsilon_{X,Y}$, and is closed under the tensor product, composition and currying.

We note from equations (14) and (15) that the projections $p_{X,Y}$ and $q_{X,Y}$ are logical and that the class of logical maps is closed under pairing. Furthermore, in the light of Proposition 12, the coherence maps $a_{X,Y,Z}$ and $c_{X,Y}$ can be expressed in terms of projection and pairing, so they too are logical. The same holds for the inverse maps $a^{-1}_{X,Y,Z}$, $c^{-1}_{X,Y}$ and r^{-1}_X.

Example 18. The class of singleton arrows in $\mathscr{A}_\Gamma = \wp_{\mathrm{fin}}(\mathscr{C}_\Gamma)$ has the closure property of Definition 17, so the logical maps form a subclass of these. In fact, they form a strict subclass, since the morphisms generated by the definition contain no free variables from Γ. Recall from Example 9 that the singleton arrows in $\mathscr{A}_\Gamma$ are selected. It follows, by Proposition 12, that the logical maps form a cartesian closed subcategory of $\mathscr{A}_\Gamma$. We conclude that the logical morphisms in $\mathscr{A}_\Gamma$ are the singleton arrows $(x \mapsto t)$ for which the normal form of t contains no free variable other than x. This class has the required closure property and the equivalence between the simply typed λ-calculus and cartesian closed categories tells us that the combinators given are sufficient to generate all such arrows.

Proofs differ from other arguments in that they establish their conclusion outright, so they should always be assigned the maximum confidence value. One might have thought that the selected maps fulfill this role. However, a logical map need not be selected: for example, ε is not selected in **Rel**. Conversely, not every selected map can be considered as a logical proof. In the case of $\mathscr{A}_\Gamma$, every singleton is selected, yet it may contain free variables in Γ.

Definition 19. If $\mathscr{A}$ is an evidential closed category, an *interpretation* of $(\&, \Rightarrow)$ minimal logic in $\mathscr{A}$ is a function $[\![-]\!]$ mapping propositions to objects of $\mathscr{A}$ such that

$$\begin{array}{rcl} [\![\text{true}]\!] & = & I \\ [\![\phi \,\&\, \psi]\!] & = & [\![\phi]\!] \otimes [\![\psi]\!] \\ [\![\phi \Rightarrow \psi]\!] & = & [\,[\![\phi]\!], [\![\psi]\!]\,]. \end{array} \tag{26}$$

We can assert the soundness of minimal logic with respect to such interpretations as follows.

Proposition 20. If $[\![-]\!]$ is an interpretation of $(\&, \Rightarrow)$ minimal logic in $\mathscr{A}$, and ϕ and ψ are propositions such that $\phi \vdash \psi$ is derivable, then there exists a logical morphism from $[\![\phi]\!]$ to $[\![\psi]\!]$ in $\mathscr{A}$.

Proof. A routine induction on the length of derivation. □

Using the Curry–Howard isomorphism, there is a canonical interpretation $[\![-]\!]_0$ of $(\&, \Rightarrow)$ minimal logic in $\mathscr{A}_\Gamma$ that gives rise to the following completeness property.

Proposition 21. If there exists a logical morphism $l : [\![\phi]\!]_0 \to [\![\psi]\!]_0$ in $\mathscr{A}_\Gamma$, then $\phi \vdash \psi$ is derivable in the $(\&, \Rightarrow)$ fragment of minimal logic.

Proof. A logical morphism $[\![\phi]\!]_0 \to [\![\psi]\!]_0$ in $\mathscr{A}_\Gamma$ yields a λ-term t in normal form with free variable x such that $x : [\![\phi]\!]_0 \vdash t : [\![\psi]\!]_0$ (the context Γ is irrelevant since t contains no free variables other than x). The Curry–Howard isomorphism associates a natural deduction proof of $\phi \vdash \psi$ to such a term. □

Thus, restricting ourselves to the logical maps of an evidential closed category, we can provide a proof theoretic semantics for $(\&, \Rightarrow)$ minimal logic with respect to which it is both sound and complete.

3. Confidence measures

Given an evidential closed category $\mathscr{A}$, we would like to regard it as a system of uncertain inference whose objects are propositions and whose morphisms are arguments. However, to do this, we need some way of assessing the support that an arrow $f : A \to B$ lends to its codomain. We measure our *confidence* in the arrows of $\mathscr{A}$ by mapping them to the elements of a suitable monoid $\mathscr{M}$ in **SLat** via an indexed family of maps $c_{A,B} : \mathscr{A}(A, B) \to \mathscr{M}$. The elements of $\mathscr{M}$ can be viewed as propositions in a fragment of *affine* logic (linear logic + weakening), in which case, $c_{A,B}(f)$ is the proposition 'f is a convincing argument'. Since 0 is never a convincing argument and $f \vee g$ is a convincing argument if and only if one of f or g is convincing, it is clear that the maps $c_{A,B}$ should be semilattice homomorphisms.

Definition 22. Let $\mathscr{A}$ be an evidential closed category and $\langle \mathscr{M}, \cdot, \top \rangle$ be a commutative monoid in $\langle \mathbf{SLat}, \underline{\otimes}, I \rangle$ for which the identity $\top$ is also a top element. Then a *confidence measure* for $\mathscr{A}$ valued in $\mathscr{M}$ is a family $c_{A,B} : \mathscr{A}(A, B) \to \mathscr{M}$ of maps in **SLat** such that

1 $c(1_X) = c(l_X) = c(r_X) = c(\Delta_X) = c(t_X) = \top$,
2 $c(g) \cdot c(f) \leq c(gf)$ whenever f and g are composable,
3 $c(f \otimes 1_X) = c(f)$,
4 $c(\Lambda f) = c(f)$ for all $f : X \otimes Y \to Z$.

The important properties of a confidence measure are summarized by the following lemma.

Lemma 23. Let $\mathscr{A}$ be an evidential closed category and $c : \mathscr{A} \to \mathscr{M}$ be a confidence measure on $\mathscr{A}$. Then

1 $c(l) = \top$ whenever $l : A \to B$ is logical,
2 if l_1 and l_2 are logical morphisms, then $c(f) \leq c(l_1 f l_2)$,
3 if k_1 and k_2 are logical isomorphisms with logical inverses, then $c(f) = c(k_1 f k_2)$,
4 $c(f) \cdot c(g) \leq c(f \otimes g) \leq c(g)$,
5 $c(f) \cdot c(g) \leq c([f, g]) \leq c(g)$.

Proof.

1–3 Follow easily from the definitions.
4 The first part is immediate. For the second, note that $c(f \otimes g) \leq c(tf \otimes g) \leq c(t \otimes g)$. Now observe that the following diagram commutes

$$\begin{tikzcd}
X \otimes Y \arrow[r, "\Delta \otimes 1"] \arrow[d, "1 \otimes g"'] & (X \otimes X) \otimes Y \arrow[r, "a", "\cong"'] \arrow[d, "(1 \otimes t) \otimes g"'] & X \otimes (X \otimes Y) \arrow[d, "1 \otimes (t \otimes g)"] \\
X \otimes Z & (X \otimes I) \otimes Z \arrow[l, "r \otimes 1"] \arrow[r, "a", "\cong"'] & X \otimes (I \otimes Z)
\end{tikzcd} \tag{27}$$

and hence that $c(t \otimes g) = c((1 \otimes t) \otimes g) \leq c(1 \otimes g) = c(g)$.
5 Again the first part is immediate. For the second, we check the commutativity of the following diagram

$$\begin{tikzcd}
U \otimes Y \arrow[r, "g \otimes 1"] \arrow[d, "k \otimes 1"'] \arrow[dr, phantom, "\leq"] & V \otimes Y \arrow[r, "1"] \arrow[d, "k \otimes 1"'] \arrow[dr, phantom, "="] & V \otimes Y \\
{[X, U] \otimes Y} \arrow[r, "{[f, g] \otimes 1}"'] & {[Y, V] \otimes Y} \arrow[r, "a(1 \otimes \Delta)"'] & {([Y, V] \otimes Y) \otimes Y} \arrow[u, "\varepsilon \otimes 1"']
\end{tikzcd} \tag{28}$$

where $k_{X,U} : U \to [X, U]$ is the currying of the first projection. It follows that $c([f, g]) = c([f, g] \otimes 1) \leq c(g \otimes 1) = c(g)$. □

Example 24. Let $\{X_i\}_{i \in S}$ be a family of nonempty sets and let $\mathscr{R}$ be the full subcategory of **Rel** whose objects are finite products of the X_i.

Suppose that for each $i \in S$ we have a specific choice of element $x_i \in X_i$. We extend the notion of chosen element to each of the objects in $\mathscr{R}$ as follows:

— the unique element $* \in I$ is chosen;
— if $u \in U$ and $v \in V$ are chosen elements, so is $\langle u, v \rangle \in U \times V$.

If u and v are the chosen elements of U and V, respectively, we define a map $c_{U,V}$: $\mathscr{R}(U,V) \to \mathbf{2}$ as follows:

$$c_{U,V}(F) = \begin{cases} 1 & \text{if } \langle u,v\rangle \in F \\ 0 & \text{otherwise.} \end{cases} \tag{29}$$

It is easy to verify that the maps $c_{U,V}$ give a confidence measure on $\mathscr{R}$. Furthermore, since the definition of c is parametric in the choice of x_i, we can define a new family of maps $c'_{U,V} : \mathscr{R} \to \wp(\Pi_{i\in S}X_i)$ by taking $c'(F)$ to be the set of choices $\langle x_i\rangle$ for which $c(F) = 1$. This too is a confidence measure on $\mathscr{R}$.

Example 25. Let $\Gamma = \{v_1 : A_1, \dots, v_m : A_m\}$ be a context in the simply typed λ-calculus and let $\mathscr{D}$ be the free distributive lattice generated by the variables v_i. We define a family of maps $c_{A,B} : \mathscr{A}_\Gamma(A,B) \to \mathscr{D}$ as follows:

$$c(x \mapsto t_1, \dots, t_n) = \bigvee_{1\le j\le n} \bigwedge \left\{ v_i \;\middle|\; \begin{array}{l} v_i : A_i \in \Gamma \text{ and } v_i \text{ occurs as a free} \\ \text{variable in the normal form of } t_j \end{array} \right\}. \tag{30}$$

The logical maps are those arrows $(x \mapsto t)$ such that the normal form of t contains no free variables in Γ. It follows that these are mapped to $\top$. A variable v_i can only appear in the normal form of $u[t/y]$ if it appears in the normal form of either t or u. Thus

$$c(y \mapsto u) \wedge c(x \mapsto t) \le c(x \mapsto u[t/y]), \tag{31}$$

and the same holds for general arrows in $\mathscr{A}_\Gamma$ by distributivity. Since neither currying nor tensoring with an identity affecs the variables of Γ that occur in a morphism, we conclude that the maps $c_{A,B}$ determine a confidence measure on $\mathscr{A}_\Gamma$.

Note that the inequality (31) is often strict, because the substitution may introduce new redexes and subsequent reduction could then eliminate some of the free variables. This observation is central to the theory of confidence measures.

Example 25 forms the basis for many more important examples via the following lemma.

Lemma 26. Let $\mathscr{M}_1$ and $\mathscr{M}_2$ be commutative monoids in $\langle \mathbf{SLat}, \underline{\otimes}, I\rangle$ for which the identities are top, and suppose that there is a lax **SLat**-functor $F : \mathscr{M}_1 \to \mathscr{M}_2$. Then any confidence measure valued in $\mathscr{M}_1$ can be extended to one valued in $\mathscr{M}_2$ by composing with F.

Example 27. Let $c : \mathscr{A}_\Gamma \to \mathscr{D}$ be the confidence measure of Example 25. Given a basic assignment s of strengths in the interval $[0,1]$ to the variables in Γ, we can extend this to a lattice homomorphism $l : \mathscr{D} \to \langle [0,1], \min, 1\rangle$. The composite $lc : \mathscr{A}_\Gamma \to [0,1]$ is a confidence measure that assesses the strength of an argument as the strength of its weakest link.

Example 28. As a modification of the last example, we can consider the lax functor $w : \mathscr{D} \to \langle [0,1], *, 1\rangle$ induced by mapping a conjunction of distinct variables to the product of their strengths. To do this, we note that any element x of $\mathscr{D}$ can be written in a disjunctive normal form

$$x = \bigvee_{1\le i\le k} \bigwedge_{1\le j\le m_i} v_{i,j} \tag{32}$$

such that none of the conjuncts contains a variable repeated and none of the disjuncts subsumes any other. This normal form is unique up to permutation, and we can therefore define w on x by

$$w(x) = \max_{1 \le i \le k} \prod_{1 \le j \le m_i} s(v_{i,j}). \tag{33}$$

It is easy to verify that $w : \mathscr{D} \to \langle [0,1], *, 1 \rangle$ is a lax **SLat**-functor, and so the composite $wc : \mathscr{A}_\Gamma \to \langle [0,1], *, 1 \rangle$ is a confidence measure.
By applying the order reversing isomorphism $r \mapsto -\log r$ between the interval $[0,1]$ under multiplication and $[0,\infty]$ under addition, we can regard the number $s(v)$ assigned to a variable v as a measure of the work one would have to expend in order to convince the listener of the basic argument step v. The confidence $wc(f)$ assigned to an argument $f \in \mathscr{A}_\Gamma(A, B)$ is thus a measure of the work one would have to expend to convince the listener of the compound argument f.

3.1. *The probability of provability*

The Dempster–Shafer theory of evidence (Shafer 1976) is perhaps the most important of the numerical uncertainty calculi that have arisen in contention with the classical theory of probability. Shafer's objection to probability is much the same as the intuitionists' objection to classical logic: that it assumes a priori that every proposition ϕ has a value 'true' or 'false' without regard to our knowledge of ϕ. Thus, the probabilities of ϕ and $\neg\phi$ must sum to 1 even though we may have no evidence to support either. In contrast, the Dempster–Shafer theory assigns a belief to each proposition such that $Bel(0) = 0$ and $Bel(\top) = 1$, but the usual additive rule is weakened to an inequality

$$Bel(\phi) + Bel(\psi) \le Bel(\phi \wedge \psi) + Bel(\phi \vee \psi). \tag{34}$$

Let ψ be a proposition in classical logic. The belief assigned to ψ is calculated from the 'belief masses' assigned to the initial premises Φ (representing the available evidence) by the repeated use of *Dempster's rule*. The easiest way to understand this is via the 'probability of provability' (Wilson 1989; Clarke and Wilson 1991). Imagine a series of trials, each one consisting of the following steps:

1 Pick a subset Φ' of Φ as follows:

   (a) take each of the elements of Φ in turn and either select it or reject it with a probability corresponding to its belief mass;
   (b) determine the consistency of the resulting set – if it is inconsistent, go back to 1a and choose again.

2 Decide whether $\Phi' \vdash \psi$.

The trial succeeds if ψ is derivable as a consequence of Φ' and fails otherwise. As the number of trials increases, the proportion of successful trials tends to $Bel(\psi)$.

As noted in Wilson (1989), this procedure can be carried out in any logic with a decidable consequence relation, and this gives rise to a generalisation of Dempster–Shafer belief for nonclassical logics. In the case of $(\&, \Rightarrow)$ minimal logic, there is no given notion of contradiction, so the procedure can be simplified by eliminating the consistency check 1b.

This generalised notion of Dempster–Shafer belief can be related to argumentation by considering the probability that a particular argument will succeed in establishing its conclusion. If Γ is a context, $\mathscr{A}_\Gamma$ is the category of arguments constructed from the assumptions in Γ. The selection of a subset Γ' of Γ divides the arrows of $\mathscr{A}_\Gamma$ into two classes. An arrow $(x \mapsto t_1, \ldots, t_n)$ is said to *succeed* if there is an $i \in \{1, \ldots, n\}$ such that all the free variables in the normal form of t_i apart from x are contained in Γ'. An arrow is said to *fail* if it does not succeed. Given some probability distribution for the choice of Γ', the probability that an argument will succeed is the probability attached to its image under the confidence measure $c : \mathscr{A}_\Gamma \to \mathscr{D}$ of Example 25. The probability of provability for a specific object A of $\mathscr{A}$ is the supremum of the probabilities attached to each of the arrows in $\mathscr{A}_\Gamma(I, A)$.

We can extend these ideas to the general case.

Definition 29. A *probability valuation* on a distributive lattice $\mathscr{D}$ is a monotone function $p : \mathscr{D} \to [0, 1]$ such that

1 $p(0) = 0$,
2 $p(\top) = 1$,
3 $p(x) + p(y) = p(x \wedge y) + p(x \vee y)$ for all x and y in $\mathscr{D}$.

Theorem 30. Let $\mathscr{A}$ be an evidential closed category and $[\![-]\!]$ be an interpretation of $(\&, \Rightarrow)$ minimal logic in $\mathscr{A}$. Suppose that $p : \mathscr{D} \to [0, 1]$ is a probability valuation on a distributive lattice $\mathscr{D}$ and that $c : \mathscr{A} \to \mathscr{D}$ is a confidence measure on $\mathscr{A}$. Let $S : obj(\mathscr{A}) \to [0, 1]$ be the 'support' function defined by

$$S(\phi) = \bigvee \{pc(a) \mid a \in \mathscr{A}(I, [\![\phi]\!])\}. \tag{35}$$

Then

1 $S(\text{True}) = 1$,
2 if $\phi \vdash \psi$ is derivable in minimal logic then $S(\phi) \leq S(\psi)$,
3 if $\phi_1 \vdash \psi$ and $\phi_2 \vdash \psi$ are derivable in minimal logic, then

$$S(\phi_1) + S(\phi_2) \leq S(\phi_1 \,\&\, \phi_2) + S(\psi). \tag{36}$$

Proof.

1 Note that the identity on $I = [\![\text{True}]\!]$ is a logical morphism, and thus maps to 1 under pc.
2 If $\phi \vdash \psi$, there is a logical morphism $l : [\![\phi]\!] \to [\![\psi]\!]$. It follows that

$$S(\phi) = \bigvee_a pc(a) \leq \bigvee_a pc(la) \leq S(\psi). \tag{37}$$

3 Since addition on the reals distributes over suprema, we have

$$S(\phi_1) + S(\phi_2) = \bigvee_{a_1, a_2} pc(a_1) + pc(a_2), \tag{38}$$

where a_1 and a_2 range over $\mathscr{A}(I, [\![\phi_1]\!])$ and $\mathscr{A}(I, [\![\phi_2]\!])$, respectively. By the definition of a probability valuation, this is equal to

$$\bigvee_{a_1,a_2} p(c(a_1) \wedge c(a_2)) + p(c(a_1) \vee c(a_2)). \tag{39}$$

Now, by Lemma 23, we have

$$p(c(a_1) \wedge c(a_2)) = pc(\langle a_1, a_2 \rangle) \leq S(\phi_1 \,\&\, \phi_2), \tag{40}$$

and, since there are logical morphisms $l_1 : [\![\phi_1]\!] \to [\![\psi]\!]$ and $l_2 : [\![\phi_2]\!] \to [\![\psi]\!]$,

$$p(c(a_1) \vee c(a_2)) \leq pc(l_1 a_1 \vee l_2 a_2) \leq S(\psi), \tag{41}$$

whence the result follows. $\square$

4. Conclusion

A criticism that is often levelled at techniques developed in artificial intelligence is that they are ad hoc attempts to solve a problem without a firm understanding of the underlying principles. Although this might initially appear to be the case with the form of argumentation implemented in the Oxford System of Medicine, we have shown, by providing a clear mathematical semantics, that the intuitions behind it actually make good logical sense. In fact, weakening the intuitionists idea of 'proof' so that it becomes subject to some measure of uncertainty is philosophically well motivated in its own right, and the semantics developed here could have just as easily have arisen from purely theoretical considerations.

A clear semantics for argumentation allows us to verify the soundness of the OSM. We are able to say that the behaviour of the OSM is justified in that it correctly computes the objects of a well-defined mathematical theory. The current OSM is restricted to Horn clause logic, but this can easily be extended to the $(\&, \Rightarrow)$ fragment of minimal logic considered here. There is an efficient backward-chaining algorithm for proof search in this logic, which, given a context Γ of the simply typed λ-calculus, allows us to find all the terms of a given type, and hence explicitly calculate the arrows of $\mathscr{A}_\Gamma$. This forms the basis of an 'argumentation theorem prover' that is sound and complete with respect to the semantics. The theorem prover comes equipped with various modules, which calculate the different confidence measures on $\mathscr{A}_\Gamma$, and these components are now being used in a rational reconstruction of the OSM.

There are several obvious extensions of the theory presented here. One can imagine a form of argumentation for any fragment of logic with a well-defined theory of proof. This can be automated provided that proof search is terminating and the equivalence of proofs is decidable. For example, if one extends $(\&, \Rightarrow)$ minimal logic by a modal operator $\diamond$ of possibility, then arguments can be represented as finite sets of terms in (a suitable modification of) the λ_c-calculus (Moggi 1989). The categorical semantics of such arguments should then correspond to an evidential closed category equipped with an **SLat**-enriched monad. Other extensions require more thought. For instance, it is difficult to say how best to treat negation. The standard approach, used in both minimal and intuitionistic logic, is to introduce a distinguished proposition $\perp$ representing 'contradiction' or 'falsehood' and define $\neg\phi$ to be $\phi \Rightarrow \perp$. However, in intuitionistic logic, $\perp$ entails every other proposition, and in minimal logic, it entails all the negated ones. In argumentation, neither of these seems satisfactory. Arguments for $\perp$ represent conflicting evidence and it is not clear that this provides positive support for anything. One could perhaps consider other encodings of negation (for example, using modal logic to represent $\neg\phi$ as $\Box(\phi \Rightarrow \perp)$), but it seems wiser to consider negation as something external to argumentation for the moment.

Although the theory of argumentation has been developed in response to the needs of the OSM, it is clear that the ideas have application far beyond this. In some ways it might be seen as a general paradigm for practical reasoning rather than a specific technique. There are aspects of it that touch upon the ideas of default reasoning, belief revision, informal argument and even linguistics. For example, consider the following algorithm for belief revision based upon argumentation.

Let Γ be some context representing our current belief state and suppose that we have just discovered that some proposition ϕ that is a consequence of Γ is definitely false. Let $(x \mapsto t_1, t_2, \ldots, t_n) : I \to [\![\phi]\!]$ be an arrow in $\mathscr{A}_\Gamma$ representing the evidence for ϕ. We must revise our beliefs so that each of the arguments $t_1, t_2, \ldots, t_n$ is retracted. Thus we identify a number of *culprit sets*, subsets X of Γ such that each term t_i contains a variable in X. Belief revision is achieved by choosing a suitable culprit set and deleting its elements from Γ. If we have a confidence measure on $\mathscr{A}_\Gamma$, or some other notion of strength, then this can be used to guide the choice of culprit set. We would aim to delete the set with the lowest overall strength attached.

This algorithm was originally proposed by Crouch and Pulman (Crouch and Pulman 1991) without any reference to argumentation; for instance, they related the choice of culprit set to a level of *epistemic entrenchment*. Stating it in the form above makes it clear just how natural the algorithm is in terms of argumentation, but also begins to relate argumentation to more standard notions in practical reasoning.

We have established a link between argumentation and the Dempster–Shafer theory of evidence. It might also be interesting to consider its relation to other uncertainty calculi, such as possibility theory (Dubois and Prade 1988) and classical probability. Furthermore, one can imagine using arguments to establish *values* (what is worth doing) and *obligations* (what ought to be done) as well as beliefs (what is likely to be true). It may be that there are connections to the theory of utility (Lindley 1985) and deontic logic (Aqvist 84).

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